

CUSTER, SD, USA

WMO: 726514

Lat: 43.733N

Lon: 103.611W

Elev: 5545

StdP: 11.98

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94032

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.5	-1.6	-17.3	2.6	-1.4	-11.4	3.6	3.0	24.2	34.8	20.7	33.3	3.6	270	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	87.7	59.4	84.1	59.2	81.2	58.8	64.4	77.0	62.9	76.0	61.5	75.0	9.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	97.8	69.1	58.6	90.4	68.0	56.7	84.4	66.9	32.5	77.0	31.2	75.9	30.1	75.0	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	17.6	15.7	-17.5	92.8	4.1	2.3	-20.5	94.4	-22.9	95.8	-25.2	97.0	-28.2	98.7		
(5)			-18.0	67.8	4.1	1.1	-20.9	68.7	-23.3	69.3	-25.6	70.0	-28.5	70.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	25.2	23.8	32.4	39.2	48.0	58.6	66.6	64.6	56.0	42.8	33.0	24.9		
	DBStd	17.80	11.13	12.27	10.92	9.24	8.60	7.38	5.86	5.96	8.79	9.95	11.10	11.22		
	HDD50	4151	769	734	549	340	142	14	0	1	43	264	515	780		
	HDD65	8226	1234	1154	1010	775	527	213	52	80	287	689	961	1244		
	CDD50	1608	0	0	4	15	81	273	516	453	222	40	4	0		
	CDD65	210	0	0	0	0	1	23	102	67	17	0	0	0		
	CDH74	3070	0	0	0	2	51	415	1308	945	338	11	0	0		
	CDH80	871	0	0	0	0	6	106	430	265	64	0	0	0		
	WSAvg	6.8	7.2	7.1	7.0	7.2	7.2	6.7	6.2	6.0	6.2	6.6	6.8	7.1		
	PrecAvg	18.0	0.4	0.5	0.9	1.9	3.1	3.3	3.1	2.0	1.5	0.9	0.6	0.6		
(15) Precipitation	PrecMax	23.0	0.9	2.0	2.1	3.9	8.8	6.4	7.6	4.0	4.8	2.5	1.7	1.7		
	PrecMin	11.9	0.0	0.0	0.0	0.3	0.1	0.7	0.4	0.5	0.1	0.1	0.0	0.0		
	PrecStd	3.4	0.2	0.5	0.5	1.0	2.0	1.7	1.7	1.0	1.3	0.7	0.4	0.4		
	PrecStd	3.4	0.2	0.5	0.5	1.0	2.0	1.7	1.7	1.0	1.3	0.7	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	55.9	65.9	72.2	79.9	88.6	92.8	89.8	85.6	75.2	65.7	53.9		
		MCWB	38.4	40.2	44.8	48.5	55.0	58.6	60.0	58.2	56.9	50.5	44.4	38.1		
	2%	DB	48.0	50.2	60.0	66.5	74.1	82.8	88.7	86.2	81.5	70.0	59.6	48.0		
		MCWB	35.6	36.9	42.6	46.8	53.6	58.6	60.5	58.6	55.7	49.1	42.3	36.0		
	5%	DB	44.1	45.7	54.7	61.9	69.5	79.1	84.7	82.9	78.3	64.6	54.3	44.2		
		MCWB	33.6	34.9	40.1	45.1	52.5	58.5	60.3	59.0	54.8	46.7	39.8	33.4		
	10%	DB	40.6	41.0	50.0	56.2	64.4	74.6	81.4	79.7	73.9	59.5	49.4	39.9		
		MCWB	31.7	32.2	37.8	42.7	50.8	57.2	60.1	58.8	53.5	44.5	37.5	31.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	41.2	46.8	50.8	59.3	65.8	67.0	65.5	62.5	53.7	45.6	39.3		
		MCDB	52.0	53.6	62.4	66.5	72.1	75.5	78.8	78.3	75.6	69.4	63.2	50.6		
	2%	WB	36.4	37.9	43.4	48.1	56.3	62.4	65.0	63.7	58.9	50.7	42.5	36.7		
		MCDB	46.3	48.6	58.3	62.6	68.2	74.1	77.7	77.1	74.2	64.4	57.5	47.1		
	5%	WB	34.3	35.2	40.8	45.9	54.0	60.4	63.5	62.2	57.0	48.3	40.3	34.1		
		MCDB	43.1	44.9	53.8	60.1	66.0	73.9	76.7	75.2	72.6	61.7	53.3	42.9		
	10%	WB	32.2	32.6	38.2	43.6	51.5	58.7	62.0	60.8	55.0	45.9	37.8	32.0		
		MCDB	40.3	40.6	49.0	55.3	63.2	72.3	76.0	73.6	70.2	58.1	48.8	40.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	19.7	20.7	21.0	21.0	23.3	24.9	25.1	25.0	21.2	19.9	18.7		
		MCDBR	21.3	22.0	27.3	30.7	28.7	30.3	30.3	30.2	30.8	29.3	25.4	20.6		
		MCWBR	12.9	13.2	14.6	15.4	13.5	13.1	11.3	11.5	12.7	14.2	13.6	12.6		
	5% WB	MCDBR	20.8	21.6	25.9	28.2	26.1	26.6	25.5	24.9	27.2	26.3	24.6	19.8		
		MCWBR	13.2	13.6	14.3	15.1	13.6	13.3	11.7	11.8	12.2	13.8	13.9	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.235	0.243	0.258	0.284	0.296	0.310	0.330	0.327	0.295	0.268	0.244	0.233		
	taud		2.625	2.601	2.611	2.520	2.566	2.541	2.506	2.495	2.584	2.633	2.611	2.613		
	Ebn at Noon		298	312	317	312	310	304	296	294	298	297	291	288		
	Edh at Noon		19	23	26	31	31	32	33	32	27	22	19	17		
(44) All-Sky Solar Radiation	RadAvg		632	921	1298	1604	1895	2231	2246	1943	1546	1020	691	524		
	RadStd		31	53	103	102	180	157	80	75	92	91	49	38		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page